A constant-map counterexample to a proposed chaos criterion

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Abstract

The inequality $|\alpha + 1| > |\beta|$ does not imply the existence of chaotic solutions of $z_{n+1} = (\alpha + \alpha z_n + \beta z_{n-1})/(1 + z_n)$. For $\beta = 0$ and $\alpha \neq -1$, every admissible solution is exactly constant after one scalar step. In particular $\alpha = i$, $\beta = 0$ disproves MathDB 331783 and the source's Conjecture 5.1.

Task 1: full counterexample and source scope

Hassan's manuscript [1], page 1, permits arbitrary complex parameters and initial values. Conjecture 5.1 on page 13 asserts that chaotic solutions exist if and only if $|\alpha + 1| > |\beta|$. The conjecture does not require $\beta \neq 0$. The source identifies chaos numerically through positive Lyapunov exponents on page 12. The example below has a constant second iterate as a state-space map, so it has no chaotic orbit under that criterion or under the usual orbit-separation criteria.

Theorem 1. *For every $\alpha \in \mathbb{C} \setminus \{-1\}$ and $\beta = 0$, each initial pair $(z_{-1}, z_0) \in \mathbb{C}^2$ with $z_0 \neq -1$ gives an everywhere-defined forward solution satisfying*

$$z_n = \alpha \qquad \text{for all } n \geq 1.$$

The strict inequality $|\alpha + 1| > |\beta|$ holds, but there are no chaotic solutions. In particular $(\alpha, \beta) = (i, 0)$ is a counterexample to the conjecture.

Proof. At every point where the original denominator is nonzero,

$$\frac{\alpha + \alpha z_n + 0 z_{n-1}}{1 + z_n} = \frac{\alpha(1 + z_n)}{1 + z_n} = \alpha.$$

Thus $z_1 = \alpha$. Since $\alpha \neq -1$, the next denominator is $1 + \alpha \neq 0$, and induction gives the asserted solution for all $n \geq 1$. No assignment at the excluded pole $z_n = -1$ is used.

For completeness, put $D = \{(x, y) \in \mathbb{C}^2 : y \neq -1\}$. The state-space map on this forward-invariant domain is

$$F(x, y) = (y, \alpha), \qquad F^2(x, y) = (\alpha, \alpha).$$

Every pair of forward orbits therefore agrees exactly after two iterations. In particular there is no persistent separation of nearby orbits and no Li–Yorke scrambled pair: the distance between any two iterated states is zero for every $n \geq 2$. The complex Jacobian is

$$DF = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad D(F^n) = 0 \quad (n \geq 2).$$

Viewed as a real four-dimensional map, the same nilpotence holds. All infinitesimal perturbations are annihilated in two steps, so no Lyapunov exponent can be positive (with the usual extended-real convention their growth rates are $-\infty$). This also directly excludes the source's numerical notion of chaos.

Finally $|\alpha + 1| > 0 = |\beta|$ because $\alpha \neq -1$. For the explicit nonreal choice $\alpha = i$, the squared left-hand side is $|1 + i|^2 = 2$, and all admissible solutions have constant tail $i, i, i, \ldots$. Thus the sufficiency direction, and hence the claimed equivalence, is false. □

The counterexample concerns every admissible initial pair at the chosen parameters, rather than merely a single fixed-point initial condition. Undefined initial data at $z_0 = -1$ do not generate solutions of the original recurrence and cannot supply a chaotic orbit.

Four proposed follow-up tasks

Task 2. Determine whether any sufficient chaos criterion depending only on $|\alpha + 1|$ and $|\beta|$ survives after imposing $\beta \neq 0$.

Task 3. Find explicit nonzero parameter regions admitting a forward-invariant domain on which all trajectories converge to one fixed point, and compare them with the proposed inequality.

Task 4. Give rigorous invariant sets and certified Lyapunov bounds for the numerical parameter pairs in the source's Table 4.

Task 5. Classify parameter pairs for which some iterate of the state-space map is constant or has lower-dimensional image.

References

[1] Sk. Sarif Hassan, *Dynamics of* $z_{n+1} = (\alpha + \alpha z_n + \beta z_{n-1})/(1 + z_n)$ *in Complex Plane*, arXiv:1511.04363v1 (2015), p. 1 and Conjecture 5.1, p. 13.